

95

CHE 312 HW #3

2.7 $q''_x = -k \frac{dT}{dx}$

a.) $q''_x = -(120 \text{ W/m} \cdot \text{K}) \cdot \left(\frac{700 - 500 \text{ K}}{0.120 \text{ m}} \right) = -2.0 \times 10^5 \text{ W}$

$\frac{dT}{dx} = \left(\frac{700 - 500 \text{ K}}{0.120 \text{ m}} \right) = 1666.67 \text{ K/m}$

b.) $q''_x = -(120 \text{ W/m} \cdot \text{K}) \left(\frac{500 - 700 \text{ K}}{0.120 \text{ m}} \right) = 2.0 \times 10^5 \text{ W}$

$\frac{dT}{dx} = \left(\frac{500 - 700 \text{ K}}{0.120 \text{ m}} \right) = -1666.67 \text{ K/m}$

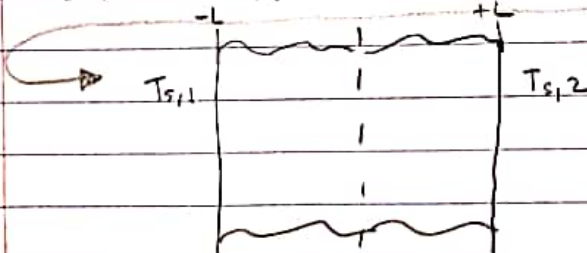
c.) $q''_x = -(120 \text{ W/m} \cdot \text{K}) \left(\frac{700 - 500 \text{ K}}{0.120 \text{ m}} \right) = -2.0 \times 10^5 \text{ W}$

$\frac{dT}{dx} = \left(\frac{700 - 500 \text{ K}}{0.120 \text{ m}} \right) = 1666.67 \text{ K/m}$

2.19 $\frac{d}{dx} \left(k \frac{dT}{dx} \right) + \dot{q} = \rho c_p \frac{dT}{dt}$ S.S.

$k \frac{d^2T}{dx^2} = -\dot{q} \Rightarrow \frac{d^2T}{dx^2} = -\frac{\dot{q}}{k}$

$\frac{d^2T(x)}{dx^2} = -\frac{\dot{q}}{k} \xrightarrow{\text{Integrate}} \frac{dT(x)}{dx} = -\frac{\dot{q}}{k} x + C_1 \xrightarrow{\text{Integrate}} T(x) = -\frac{\dot{q}}{2k} x^2 + C_1 x + C_2$



Boundary Conditions

@ $x = -L$ $T = T_{s1}$

@ $x = L$ $T = T_{s2}$

B.C. 1: $T_{s1} = -\frac{\dot{q}}{2k} L^2 - C_1 L + C_2$

B.C. 2: $T_{s2} = -\frac{\dot{q}}{2k} L^2 + C_1 L + C_2$

Add Both eqns: $T_{s1} + T_{s2} = -\frac{\dot{q}}{k} L^2 + 2C_2 \Rightarrow \frac{T_{s1} + T_{s2}}{2} = -\frac{\dot{q} L^2}{2k} + C_2$

Substitute into Original Eqn.

$T_{s2} = -\frac{\dot{q} L^2}{2k} + C_1 L + \frac{T_{s1} + T_{s2}}{2} + \frac{\dot{q} L^2}{2k}$

$C_1 = \frac{T_{s1} - T_{s2}}{2L}$ Plug C_1 & C_2 into original

$T = -\frac{\dot{q} x^2}{2k} + \left(\frac{T_{s2} - T_{s1}}{2L} \right) x + \frac{T_{s1} + T_{s2}}{2} + \frac{\dot{q} L^2}{2k}$

95
100

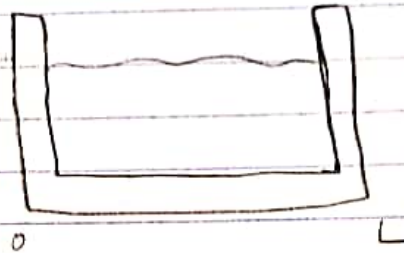
$$\begin{aligned}
 q''_{\text{cal}} &= -k \frac{\partial T}{\partial x} \\
 &= -k \frac{1}{2L} \left(\frac{qL^2}{2k} \left(1 - \frac{x^2}{L^2} \right) + \frac{T_{s2} - T_{s1}}{2} x + \frac{T_{s1} + T_{s2}}{2} \right) \\
 &= -k \left(\frac{qL}{2k} \left(-\frac{2x}{L} \right) + \frac{T_{s2} - T_{s1}}{2} \right) \\
 q''_{\text{cal}} &= qx - \frac{k}{2L} (T_{s2} - T_{s1})
 \end{aligned}$$

2.20 a.)

$$k \frac{\partial^2 T}{\partial x^2} = \rho c_p \frac{\partial T}{\partial t}$$

$$\frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

Time dependent



Boundary conditions

$$-k \frac{\partial T}{\partial x} \Big|_{x=0} = \frac{q_0}{\pi D^2/4} \quad -k \frac{\partial T}{\partial x} \Big|_{x=L} = h [T_\infty - T(0, x)]$$

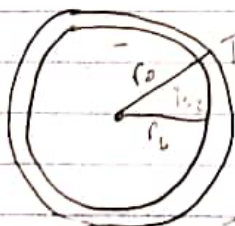
$$T(x, t) \Rightarrow T(x, 0) = T_i$$

20

b.) $T(L, t) = T(L)$ $q''_0 = -k \frac{\partial T}{\partial x} \Big|_{x=0}$ ← Boundary conditions

$$\frac{\partial^2 T}{\partial x^2} = 0 \quad \leftarrow \text{Temperature remains fixed}$$

2.40



$$T(r) = \frac{C_1}{r} + C_2$$

$$q_r = -kA \frac{\partial T}{\partial r} \leftarrow \text{Heat Rate}$$

$$q''_r = -k \frac{\partial T}{\partial r} \leftarrow \text{Heat Flux}$$

$$\frac{1}{r} \frac{\partial}{\partial r} (kr \frac{\partial T}{\partial r}) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$

Since $T(r)$ has no time dependence, this is

a Steady state problem

$$q''_r = -k \frac{\partial T}{\partial r} = -k \frac{\partial}{\partial r} \left(\frac{C_1}{r} + C_2 \right) = k \frac{C_1}{r^2}$$

This shows that r increases, the heat flux will decrease.

$$q_r = -kA \frac{\partial T}{\partial r} = -k \cdot (4\pi r^2) \cdot \left(\frac{\partial}{\partial r} \left(\frac{C_1}{r} + C_2 \right) \right)$$

$q_r = 4\pi k C_1$ → Since there is no 'r' term, the heat rate is independent of radius

25